Student worksheet

Lesson 1 · Vocabulary and observing vehicle motion · No calculations required

Name: _____ Class: _____ Date: _____

Part A — Thinking like a scientist

1. Write two ways that physics can help us understand or control a real car journey.

2. A student says, “The car is going at 20.” What important information is missing? Explain.

3. Explain, in your own words, how a physical quantity differs from the symbol used for it and its SI unit.

Part B — Eight motion quantities

4. A car drives along a winding road from school to a football stadium. Explain the difference between distance travelled and displacement.

5. Explain the difference between speed and velocity using a car example.

6. What is meant by average speed? How does it differ from an instantaneous speedometer reading?

7. A car turns a corner without speeding up or slowing down. Is it accelerating? Explain.

Part C — Scalars and vectors

Write S for scalar or V for vector, then identify whether direction is needed to describe each quantity.

Quantity	S or V?	Does direction matter?
Distance travelled		
Displacement		
Speed		
Velocity		
Average speed		
Average velocity		
Acceleration		

8. A car is travelling left. We have chosen right as positive. Is the car's speed positive, negative or zero? What about the sign of its velocity?

9. A car is located 15 m to the left of a chosen origin. Explain what is meant by its position.

Part D — At an instant or over a journey?

10. Which is an instantaneous description: (a) the car's speedometer reads 45 km/h now; (b) the car's average speed was 45 km/h on the journey? Explain.

11. A car completes a lap of a circular test track and returns to its starting point. Compare its distance travelled, displacement, average speed and average velocity.

Part E — Six car journeys

Use the Six Motion Scenarios HTML. Only describe the movement while the background is green. Do not reveal the explanations until you have predicted the answers.

Journey	What happens to the speed?	Does velocity change? Why?
A — Constant velocity		
B — One circular lap		
C — Starts from rest		
D — Brakes to a stop		
E — Constant velocity left		
F — Slows, reverses, returns		

12. For journeys C and D, the model uses the same total distance and time. What is different about how their speeds change?

13. In journey E, why is it sensible to call the velocity negative but not the speed?

14. In journeys B and F, the car returns to its starting point. What does that tell you about its displacement and average velocity?

Exit reflection

15. Complete this statement accurately: “Acceleration means a change in _____, which could happen by changing _____ or _____.”